

Dose-Based IV – Practice Solutions 1

1. **Order:** Morphine sulfate 10 mg/hr IV continuous
Available: Morphine 10 mg/5 mL (2 mg/mL)
Pump rate: 5 mL/hr

$$\frac{10 \cancel{\text{mg}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{2 \cancel{\text{mg}}}} = \frac{10}{2} = 5 \frac{\text{mL}}{\text{hr}}$$

2. **Order:** Regular insulin 20 units/hr IV continuous
Available: Insulin 100 units/50 mL NS (2 units/mL)
Pump rate: 10 mL/hr

$$\frac{20 \cancel{\text{units}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{2 \cancel{\text{units}}}} = \frac{20}{2} = 10 \frac{\text{mL}}{\text{hr}}$$

3. **Order:** Potassium chloride 10 mEq/hr IV continuous
Available: KCl 40 mEq/100 mL NS (0.4 mEq/mL)
Pump rate: 25 mL/hr

$$\frac{10 \cancel{\text{mEq}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.4 \cancel{\text{mEq}}}} = \frac{10}{0.4} = 25 \frac{\text{mL}}{\text{hr}}$$

4. **Order:** Morphine sulfate 4 mg/hr IV continuous
Available: Morphine 10 mg/5 mL NS (2 mg/mL)
Pump rate: 2 mL/hr

$$\frac{4 \cancel{\text{mg}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{2 \cancel{\text{mg}}}} = \frac{4}{2} = 2 \frac{\text{mL}}{\text{hr}}$$

5. **Order:** Heparin 1,500 units/hr IV continuous
Available: Heparin 25,000 units/500 mL NS (50 units/mL)
Pump rate: 30 mL/hr

$$\frac{1,500 \cancel{\text{units}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{50 \cancel{\text{units}}}} = \frac{1,500}{50} = 30 \frac{\text{mL}}{\text{hr}}$$

Dose-Based IV – Practice Solutions 2

- 6. Order:** Magnesium sulfate 2 g/hr IV continuous
Available: MgSO₄ 10 g/20 mL (0.5 g/mL)
Pump rate: 4 mL/hr

$$\frac{2 \cancel{\text{g}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.5 \cancel{\text{g}}}} = \frac{2}{0.5} = 4 \frac{\text{mL}}{\text{hr}}$$

- 7. Order:** Heparin 1,000 units/hr IV continuous
Available: Heparin 25,000 units/500 mL NS (50 units/mL)
Pump rate: 20 mL/hr

$$\frac{1,000 \cancel{\text{units}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{50 \cancel{\text{units}}}} = \frac{1,000}{50} = 20 \frac{\text{mL}}{\text{hr}}$$

- 8. Order:** Regular insulin 4 units/hr IV continuous
Available: Insulin 100 units/200 mL NS (0.5 units/mL)
Pump rate: 8 mL/hr

$$\frac{4 \cancel{\text{units}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.5 \cancel{\text{units}}}} = \frac{4}{0.5} = 8 \frac{\text{mL}}{\text{hr}}$$

- 9. Order:** Morphine sulfate 6 mg/hr IV continuous
Available: Morphine 15 mg/37.5 mL NS (0.4 mg/mL)
Pump rate: 15 mL/hr

$$\frac{6 \cancel{\text{mg}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.4 \cancel{\text{mg}}}} = \frac{6}{0.4} = 15 \frac{\text{mL}}{\text{hr}}$$

- 10. Order:** Potassium chloride 3 mEq/hr IV continuous
Available: KCl 20 mEq/40 mL NS (0.5 mEq/mL)
Pump rate: 6 mL/hr

$$\frac{3 \cancel{\text{mEq}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.5 \cancel{\text{mEq}}}} = \frac{3}{0.5} = 6 \frac{\text{mL}}{\text{hr}}$$